Mohammad Javad Zare

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Research Interests

- Computer Vision
- Medical Imaging
- Vision-Language Models
- Agentic AI

Education

MSc, Computer Science - Artificial Intelligence, University of Tehran, Iran — Oct 2026 – Present

BSc, Computer Science, University of Bojnord, Iran — Oct 2021 – Sep 2025

- Overall GPA: 15.41/20
- Relevant Coursework:
  - Databases (20/20)
  - Data Structures and Algorithms (18.5/20)
  - Fundamentals of Matrices and Linear Algebra (18.5/20)

Projects

Medical AI Platform (In Progress)
PyTorch, Pandas, Matplotlib, Medical Imaging

- Developing a modular platform for deep learning in medical imaging.
- Implementing a diabetic retinopathy detection pipeline for data preprocessing, model training, and evaluation.
- Designing reusable components within a shared module for use across different parts of the platform.

Credit Card Fraud Detection
Pandas, NumPy, Matplotlib, Scikit-Learn, LightGBM

- Developed ML pipelines for credit card fraud detection using feature engineering and classification models.
- Optimized decision thresholds based on precision-recall trade-offs.
- Compared ElasticNet Logistic Regression and LightGBM using F1-score, ROC-AUC, precision, and recall.
- Visualized model performance using ROC/PR curves and confusion matrices.

Blog Platform with Django & DRF
Django, DRF, PostgreSQL, Redis, Celery, Docker, Nginx

- Developed a RESTful blog platform using Django and Django REST Framework with API-based data access.
- Containerized the application with Docker Compose, orchestrating PostgreSQL, Redis, Gunicorn, and Nginx.
- Configured Celery and Redis for asynchronous background task execution, enabling non-blocking processing of long-running operations.

Skills

Programming: Python, C++
Machine Learning & AI: NumPy, Pandas, Scikit-learn, PyTorch, OpenCV
Backend & Database: Django, Django REST Framework, SQL, Redis
Tools: Git, Docker, Linux, LaTeX

Academic Activities

Computer Science Scientific Association, University of Bojnord — Sep 2023 – Sep 2025
Central Council Member (Executive Board)

- Co-initiated and co-organized the 1st UBPC Algorithmic Programming Contest.
- Co-organized the 2nd UBPC Algorithmic Programming Contest.
- Developed and deployed registration websites for three consecutive editions of the UBPC programming contest.
- Contributed to planning and coordinating events with faculty advisors and executive board members.

Certifications

Agentic AI (In Progress) — MaktabKhooneh 2026
Data Analysis and Preparation with Python — FaraDars Nov 2025
Fundamentals of Deep Learning — FaraDars Jul 2025
Machine Learning Fundamentals with Python — FaraDars Jul 2025
Advanced Django — MaktabKhooneh Oct 2023
Django 3.2 — MaktabKhooneh May 2023
Advanced Python Programming — MaktabKhooneh Oct 2022
Python Programming for Beginners — MaktabKhooneh Aug 2022

Languages

Persian: Native
English: B2